

# MACHINE LEARNING — ULTIMATE FINAL EXAM PRACTICE PACK

Comprehensive exam including conceptual, scenario-based, calculations, clustering, RL, and advanced trick questions. Show all work.

## SECTION A — CONCEPTUAL MULTIPLE CHOICE

Q1. Which is MOST likely to overfit?

- A. Linear regression with few features
- B. Deep decision tree with no pruning
- C. Logistic regression with regularization
- D. KNN with large K

Q2. Which scenario is LEAST appropriate for ML?

- A. Predicting housing prices
- B. Rule-based tax calculation
- C. Image classification
- D. Fraud detection

Q3. Increasing model complexity:

- A. Increases bias and decreases variance
- B. Decreases bias and increases variance
- C. Decreases both
- D. Increases both

Q4. As K increases in KNN:

- A. Decision boundary becomes more complex
- B. Bias increases, variance decreases
- C. Variance increases
- D. Model memorizes training data

Q5. ROC vs PR:

- A. ROC better for imbalance
- B. PR better when positive rare
- C. ROC ignores FP
- D. PR uses FPR

Q6. K-fold CV used for:

- A. Final eval
- B. Hyperparameter tuning
- C. Cleaning
- D. Feature engineering

Q7. Most flexible model:

- A. Linear regression
- B. Logistic regression
- C. KNN (K=1)
- D. Naive Bayes

Q8. Requires scaling:

- A. Tree
- B. RF
- C. KNN
- D. NB

Q9. Generative model:

- A. Logistic
- B. KNN
- C. Naive Bayes
- D. SVM

Q10. Best metric for imbalance:

- A. Accuracy

- B. PR curve
- C. MSE
- D.  $R^2$

## SECTION B — SCENARIO QUESTIONS

Q11. Large tabular data + interpretability → best model?

- A. NN
- B. RF
- C. Linear/Logistic
- D. KNN

Q12. High dimensional ( $p \gg n$ ):

- A. KNN
- B. Regularized logistic
- C. Tree
- D. DBSCAN

Q13. Small dataset:

- A. Deep NN
- B. Naive Bayes
- C. RF
- D. KNN ( $K=1$ )

Q14. Fraud detection (rare + costly FN):

- A. Max accuracy
- B. Increase threshold
- C. Lower threshold
- D. Ignore

Q15. Compare models properly:

- A. Different splits
- B. Same folds
- C. Use test repeatedly

D. Training only

Q16. Data leakage:

- A. Scale train only
- B. Scale before split
- C. CV
- D. Validation

Q17. High FN cost →

- A. Increase threshold
- B. Decrease threshold
- C. Ignore
- D. Accuracy

Q18. Train low, test high →

- A. High bias
- B. High variance
- C. Underfit
- D. Perfect

Q19. Overlapping CI →

- A. One better
- B. Other better
- C. Not significant
- D. Cannot compare

Q20. Helps KNN most:

- A. Noise
- B. Scaling
- C. Irrelevant features
- D. Remove all

## SECTION C — CALCULATIONS

Q21.  $TP=40$ ,  $FN=10$ ,  $FP=20$ ,  $TN=30 \rightarrow$  compute Accuracy, Precision, Recall, F1.

Q22. Scores  $[0.9, 0.8, 0.6, 0.4, 0.3]$ , labels  $[1, 1, 0, 1, 0]$ , threshold=0.5  $\rightarrow$  TP, FP, FN, TN.

Q23. Expected cost with  $FPR=0.2$ ,  $TPR=0.8$ ,  $p+=0.1$ ,  $c_{FP}=1$ ,  $c_{FN}=10$ .

Q24. Which model generalizes best (given train/test)? Explain.

Q25. Gradient descent:  $f(w)=w^2$ ,  $w_0=10$ ,  $\eta=0.1 \rightarrow$  next step.

## **SECTION D — CLUSTERING**

Q26. Spherical clusters → method?

Q27. Non-spherical clusters → method?

Q28. Noise/outliers → method?

Q29. Soft clustering → method?

Q30. High-dim sparse → best distance?

Q31. When spectral clustering?

Q32. KDE vs histogram difference?

## SECTION E — REINFORCEMENT LEARNING

Q33. Define state/action/reward (robot vacuum).

Q34. Policy definition.

Q35.  $V(s)$  meaning.

Q36.  $Q(s,a)$  meaning.

Q37. Q-learning vs SARSA.

Q38.  $\epsilon$ -greedy explanation.

Q39. Monte Carlo vs TD.

Q40. Bellman equation meaning.

## **SECTION G — ADVANCED / TRICK QUESTIONS**

Q41. ROC curves cross — which model better and why?

Q42. High AUC but poor precision — explain.

Q43. Show mathematically why accuracy fails for imbalance.

Q44. Design statistical test for model comparison.

Q45. Why K-means fails on circles.

Q46. Construct case DBSCAN fails.

Q47. Derive logistic regression gradient.

Q48. Show scaling impact on Euclidean distance.

Q49. Why LDA > logistic with small data.

Q50. Derive optimal threshold from cost.

Q51. MC vs TD bias/variance.

Q52. Why Q-learning is off-policy.

Q53. Compute  $V(s)$  from Bellman.

Q54. Overfitting in learning curves.

Q55. When PR AUC misleading.

Q56. Clustering vs density estimation decision.

Q57. Hierarchical vs spectral tradeoffs.

Q58. Cosine vs Euclidean example.

Q59. Why PCA hurts classification.

Q60. Regularization + bias-variance interaction.